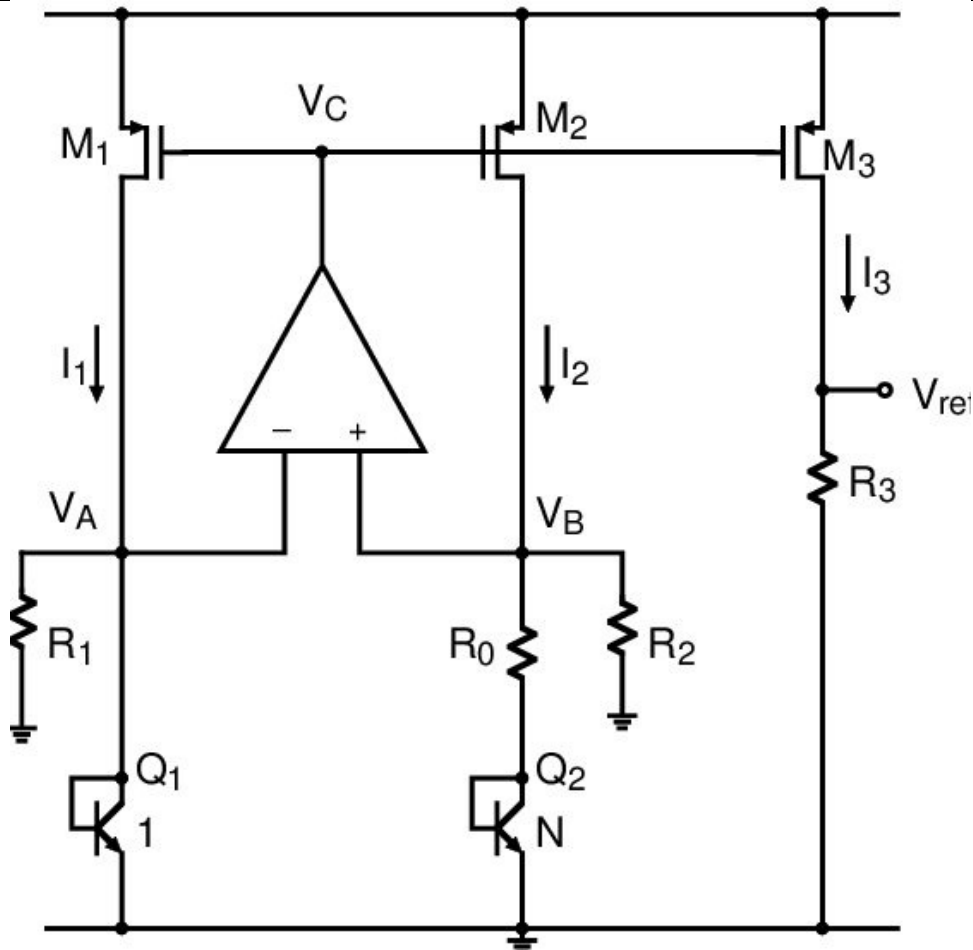


Lab 12**BGR Design**

- 1) Build a simple LDO using the folded cascode OTA (designed in lab 10)
- 2) Size R_0 so that 500 nA is flowing through it.
- 3) Size $R_1 = R_2$ so that the curve of I_2 Vs temperature is almost constant (Sweep R from 500K to 1.5M Ω).
- 4) Size R_3 so that $V_{ref} = 600$ mV.
- 5) Check the DC operating point of all MOS devices and try to consume $< 30 \mu A$ IQ (in BG and OTA).
- 6) Run Dc simulations and check V_{out} Vs Temp “-40 °C to 1.25 °C”.
- 7) Check transient response of the BGR (Ramp the supply, then plot V_{out}).
- 8) Check AC simulations and plot PSR of the BGR output (Use 1 V ac at V_{in} and plot 20 dB of V_{out}).
- 9) Check the LDO loop stability (PM and GM). Should be > 50 degrees and 12dB respectively.